Ethicode City: Gamifying Software Engineering Ethics Education

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Abstract

Software and AI are now pervasive across domains such as healthcare, finance, and transportation, with AI increasingly used for code generation and debugging. This has led to practices such as *vibe coding*, where developers prompt AI to write code and lightly edit it, often with limited verification, which can introduce subtle bugs or security flaws. Despite its importance, software engineering ethics is often taught via lectures or case studies, limiting learners' exposure to realistic ethical dilemmas.

We present *Ethicode City*, an educational game that immerses players in city-building while facing software engineering dilemmas, such as choosing between AI-assisted or manual coding, balancing efficiency with security, and maintaining quality and documentation. Unlike prior tools focused narrowly on privacy or bias, the game covers a broad spectrum of professional ethics, fostering ethical awareness, critical thinking, and decision-making. Preliminary evaluations with 8 undergraduate students indicate that *Ethicode City* effectively enhances ethical reasoning. The game is available at https://github.com/rishalab/EthicodeCity.

CCS Concepts

• **Social and professional topics → Software engineering education**; **Codes of ethics**.

Keywords

Software Engineering Ethics, Game-based Learning, AI Ethics, Gamification

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1 Introduction

AI-powered development tools, GitHub Copilot, ChatGPT, and IDE-integrated environments such as Cursor, are reshaping software engineering workflows. While these tools accelerate delivery, uncritical reliance on AI-generated code with minimal review frequently introduces security vulnerabilities and technical debt [3, 6].

Educating future engineers on these risks is challenging. Traditional pedagogical methods such as lectures and static case studies often fail to capture the tension between *delivery speed* and *ethical responsibility* [4, 5]. Students may understand ethics in theory but struggle to apply it under pressure.

To bridge this gap, we introduce **Ethicode City**, a serious game where software engineering decisions dictate the survival of a virtual city. Unlike prior games that focused narrowly on privacy [2] or algorithmic bias [1], *Ethicode City* addresses broader professional ethics, including documentation, testing, and the trade-off between AI automation and manual oversight.

2 Game Design

Ethicode City is a strategy simulation built in Unity 3D. The player assumes the role of a mayor rebuilding a collapsed city. To succeed, they must manage resources, citizen happiness, and construction projects, all mapping to software engineering concepts (see Table 1).

Game Element	SE Concept	Consequence
Manual (Coins)	Documented, tested code	Stable; earthquake-resistant
AI Factory (Tokens)	AI-generated code	Fast; hidden defect risk
Hospital	Code documentation	Resilience during crises
Bank (AI-built)	Unaudited AI output	Security breach risk
Earthquake	System stress test	Reveals deferred debt
Audit action	Code review	Mitigates AI-built risk

Table 1: Mapping of game elements to software engineering concepts.

2.1 Core Mechanics: Coins vs. Tokens

The game enforces a strict resource economy to simulate engineering trade-offs:

- **Manual Construction (Coins):** Players build factories and hospitals manually, representing code written with proper documentation and unit tests. It is slow (e.g., 200 seconds per building) but structurally stable.
- **AI-Assisted Construction (Tokens):** Players use *AI Factories* to generate tokens, representing AI-generated code. It is fast (e.g., 1 second per building) but carries hidden risks.

2.2 Ethical Scenarios

The game creates observable consequences for technical decisions:

1. The Documentation Trade-off: Players can build Hospitals, which represent code documentation. They are expensive and yield no immediate profit, but they increase city resilience. Neglecting hospitals accelerates happiness degradation during crises, simulating the long-term cost of undocumented legacy code.

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Figure 1: Gameplay progression: (1) Entry, (2) Scenario Setup, (3) The Earthquake (Testing Failure), (4) Ethical Unlocks.

2. The "Vibe Coding" Trap: Players can instantly build banks using AI tokens. However, AI-generated banks carry a high probability of *security breaches* (robberies) if not manually audited, teaching players that speed without verification leads to failure.

3. The Earthquake: Midway through the game, an earthquake occurs (Fig. 1, Panel 3), acting as a stress test. Manually constructed buildings withstand the quake; Token-rushed buildings collapse, costing massive resources to rebuild.

4. The "Well-Commented Code" Challenge: Factories operating on undocumented logic run at only 50% efficiency, reducing token output. Players must complete a *Debugging Mini-game*, annotating code snippets with meaningful comments to restore full productivity, illustrating how documentation directly impacts system performance.

5. Code Testing and Quality: As players progress, ethical badges representing practices such as *Code Testing and Quality* appear on the HUD. Collecting these badges earns reward points that unlock additional building types, reinforcing that quality assurance is an ongoing discipline rather than a one-time event.

While the consequences may appear to penalize AI-assisted construction, the intent is not to portray AI usage as unethical per se. Instead, *Ethicode City* highlights the ethical risks of uncritical AI reliance without adequate testing, documentation, or human oversight. Players can successfully leverage AI-assisted construction when complemented with manual audits and resilience-building actions, reinforcing *responsible* AI use rather than AI avoidance.

3 Implementation

The game is developed in Unity 6 (C#) using a modular scene-based architecture. A persistent *GameManager* singleton tracks global state (coins, tokens, happiness, AI dependency score), while *BuildingControllers* manage per-structure stability and breach probability. A *City Architect* NPC guides players through debugging mini-games, implemented as additively-loaded Unity scenes for independent testability. Each mini-game is designed to mirror real-world engineering decisions, ensuring that in-game challenges remain pedagogically grounded. The AI dependency metric (ratio of Token-built to total buildings) is displayed on a real-time HUD overlay, providing immediate feedback on engineering choices. This visibility encourages players to reflect on their reliance on AI throughout gameplay. Source code: https://github.com/rishalab/EthicodeCity.

4 Preliminary Evaluation

We conducted a pilot study with 8 undergraduate computer science students to assess usability and learning outcomes. We employed a survey based on the Technology Acceptance Model (TAM), adapted for educational games.

4.1 Results

Participants rated the game on a 5-point Likert scale (1=Strongly Disagree, 5=Strongly Agree). Results were promising:

- **Ethical Awareness ($M = 4.38$, $SD = 0.52$):** Players reported increased awareness of risks associated with code generation and documentation.
- **Engagement ($M = 4.38$, $SD = 0.74$):** Participants found the gamified approach more engaging than case studies.
- **Replay Value ($M = 4.75$, $SD = 0.46$):** The strongest result, indicating students were motivated to explore different strategies.

Qualitative feedback highlighted that the *Earthquake* event was a *lightbulb moment* regarding the importance of software testing. However, some users noted that the UI for the mini-games needed refinement. Given the small sample size ($n = 8$), these results should be interpreted as indicative rather than conclusive; a larger-scale study is planned.

5 Conclusion and Future Work

Ethicode City translates abstract software engineering ethics into tangible gameplay mechanics. By forcing players to choose between the immediate gratification of AI generation and the long-term stability of disciplined engineering, we create a safe environment for failure. The game is freely available and designed to complement existing software engineering courses as a hands-on ethical reasoning tool. Future work involves expanding the scenarios to include data privacy legislation and multiplayer modes to simulate team dynamics and shared responsibility, areas where ethical tensions emerge from collective, not individual, decisions.

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